ADVANCED GAN PROJECT

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Table of Contents

Dataset used 2
Related work: 2
Block diagram 3
Algorithms 3
Implementation 3
Results 4
Conclusion 5
References 6
Appendices 7
Appendix I: Model output after 25 epochs with 64 embedding space and doubled convolutional filter 7
Appendix II: Model loss curve after 25 epochs with 64 embedding space and doubled convolutional filter 7
Appendix III: Discriminator Block Diagram 7
Appendix IV: Generator Block Diagram 8

Dataset used

The dataset used for training the Conditional Generative Adversarial Network comes from the IMDB-WIKI dataset. This dataset was developed by Rothe et. al. in *IMDB-WIKI – 500k+ face images with age and gender labels.* The entire dataset contains 523,051 images from 20,284 celebrities which are sourced from IMDB and Wikipedia. The age of each celebrity is calculated as the difference between the year the photo was taken of the individual and the celebrity's date of birth. These images were cropped using the datasets facial information from their detector with at least a 50% threshold. Due to a size constraint issue, only a partial subset of the entire dataset was used for training the Conditional Generative Adversarial Networks. This subset contains approximately 141,111 cropped images of celebrity faces.

Related work:

Conditional Generative Adversarial Networks were developed as an extension of Generative Adversarial Networks (GAN). GANs were "introduced as a novel way to train generative models" but "there is no control on modes of the data being generated" (Mirza). GANs essentially employ Game Theory to produce meaningful generative outputs. By pinning the generator against a discriminator each model attempts to improve upon itself, the generator to create better outputs that are more realistic and the discriminator to better distinguish the real outputs from the fake outputs generated by the generator. Conditional GANs "can be constructed by simply feeding the data, y, we wish to condition on to both the generator and discriminator." (Mirza). Conditional GANs were originally developed in 2014 by Mirza and Osindero in *Conditional Generative Adversarial Nets* and later Gauthier in *Conditional generative adversarial nets for convolutional face generation.* Mirza and Osindero used cGANs to produce digits for the MNIST numbers and furthermore, used cGANs to auto tag images from Flickr. Gauthier used cGANs to produce human faces based on a subset of 36 attributes from the 73 total attributes which include race, age, and emotions. As can possibly be gleaned from the nature of the cGAN, Gauthier states that there is "a correlation between strong positive increases in G performance and decreases in D performance (and vice versa)" (Gauthier).

Block diagram

See Appendix III for Discriminator and Appendix IV for Generator

Algorithms

The cGAN algorithm incorporates two models, a discriminator and a generator. The discriminator will take real inputs from the dataset and fake inputs from the generator, the discriminator will then determine if the input is real or fake. The generator will produce inputs and is incentivized to produce more realistic inputs to deceive the discriminator. As a key distinction between conditional generative adversarial networks and GANs, both the generator and discriminator will take a conditional input, in this case, age.

Implementation

The images from the dataset were resized and normalized before training the model. The ages were originally encoded by the exact age of the individual but was later encoded into 20-year bins to increase the model's ability to generalize and smooth out any disparities among the various sample sizes for specific ages. To encode the images with the age bucket, the images were concatenated with their embedded conditional input. The batch size was originally set at 32 but was later changed to 16, this decision was made in part to see if a more frequent weight update would influence the generation of more realistic images. The cGAN was created by employing a discriminator and generator. The discriminator has two models. The first model reduces the spatial dimensionality and increases the feature depth. The second model categorizes the image as real or fake. The generator increases spatial dimensionality and decreases the feature depth over the layers. The embedding size for the conditional input was set at 64D. The weights were initially set at 0.02, this decreases the likelihood of the vanishing gradient. Moreover, the models employed LeakyReLU in place of ReLU to also decrease the likelihood of the vanishing gradient. In the final generator model iteration, the number of filters in the convolutional layers were doubled with the initial layer increased from 512 to 1024. Finally, the models were trained on 25 epochs.

Results

To create a robust experiment employing conditional generative adversarial networks, the embedding size dimensions was selected to determine its effectiveness in the generation of realistic facial images. Several models were trained with different embedding spaces: 32D, 64D, and 128D, see Table I, furthermore another model trained with the 64D embedding space while doubling the convolutional filter size within the generator.

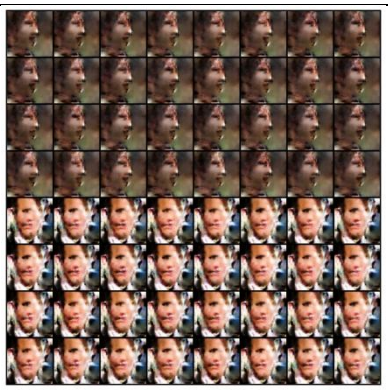
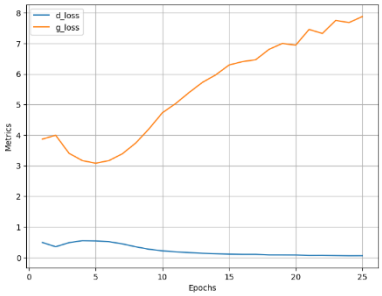
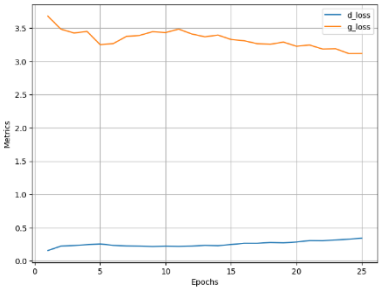
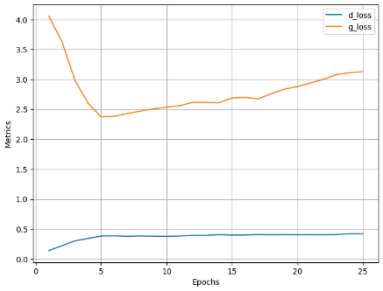
Embedding Space	***32D***	***64D***	***128D***
Generator output			
Loss over epochs			
Generator Loss	~7.9	~3.1	~3.1
Discriminator Loss	~0.05	~0.35	~0.48

Table I: Embedding Spaces Visualized

The 128D and 64D embedding spaces seem to have difficulty producing meaningful facial images, whereas the 32D embedding space appears to produce the most realistic facial images although the model does appear to have the largest generator loss in comparison to the other embedding sizes. The 32D and 128D embedding size models appear to have increasing generator losses after roughly 5 epochs. The 64D embedding size model does appear to have decreasing generator losses but does not produce very realistic facial images. Although the 64D embedding size does not lead to the production of most realistic images, the model holds the lowest generator loss with a low discriminator loss suggesting the model does produce images that can deceive a well-trained discriminator, when compared to the other models. After doubling the convolutional filter size, the generator appears to produce significantly more realistic images; see the results of embedding space 64D in Table I against Appendix I, with the larger convolutional filter. Although the larger convolutional filter seems to produce more realistic images, the loss almost doubles as well for the generator. Moreover, the larger convolutional filter model does seem to have an increasing generator loss.

Conclusion

The qualitative results indicate that increasing the convolutional filter with a slight increase to the embedding size dimension at 64D seems to generate the most realistic facial images. Whereas the quantitative results suggest the increase in the convolutional filter size within the generator increases the generator loss. In future work, an experiment that holds all else equal, increasing or decreasing the size of the convolutional filter size might offer more realistic generation of facial images more than the changing of the embedding space. For future consideration, as seen in Table I, the embedding space 64D model appears to be combining a side profile with a frontal profile in the bottom images. When examining some of the images from the IMDB dataset the images seem to feature side profiles of individuals. With a minority side profile and majority frontal profile images, the generator might have difficulty distinguishing the true features responsible for creating only the frontal profile of an individual.

References

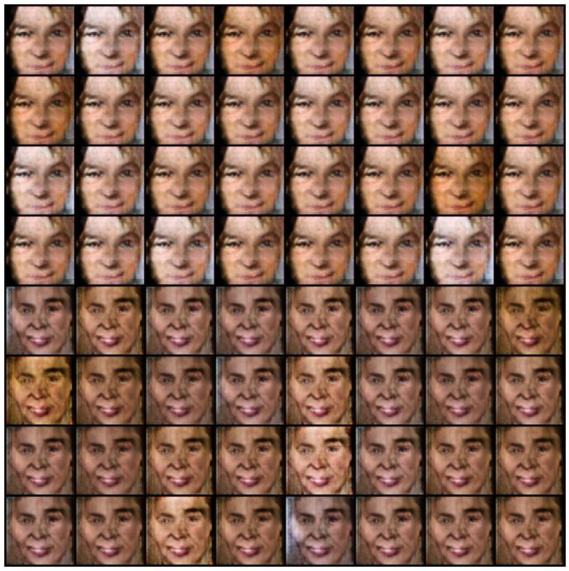
Gauthier, Jon. "Conditional generative adversarial nets for convolutional face generation." *Class project for Stanford CS231N: convolutional neural networks for visual recognition, Winter semester* 2014.5 (2014).

Mirza, Mehdi. "Conditional generative adversarial nets." arXiv preprint arXiv:1411.1784 (2014).

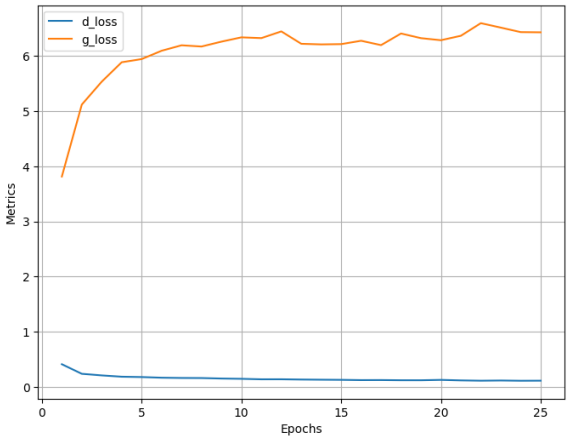
Rothe, Rasmus, et al. “Deep expectation of real and apparent age from a single image without facial landmarks.” International Journal of Computer Vision, vol. 126, 2018, pp. 144–157.

Appendices

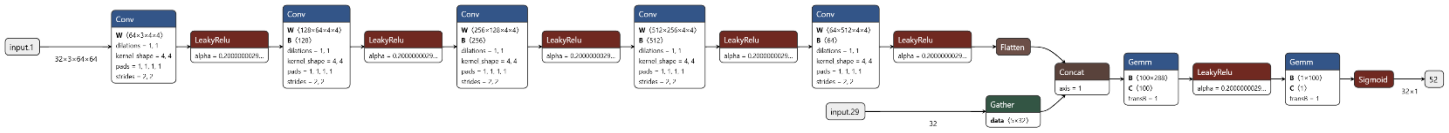
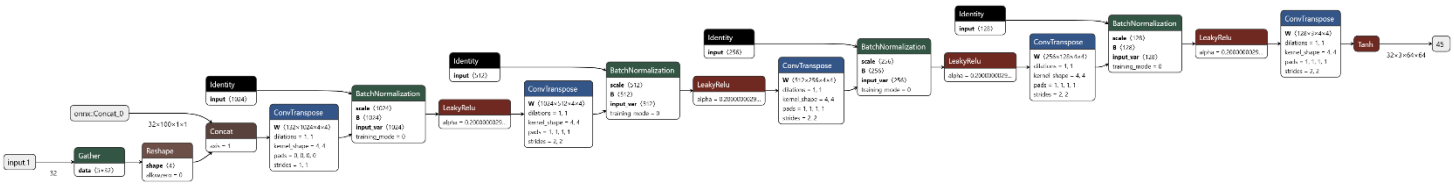
Appendix I: Model output after 25 epochs with 64 embedding space and doubled convolutional filter



Appendix II: Model loss curve after 25 epochs with 64 embedding space and doubled convolutional filter



Appendix III: Discriminator Block Diagram



Appendix IV: Generator Block Diagram